Mudit Agrawal

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Experience

- **AI/ML Intern – Carnot Research Pvt. Ltd. (IITD Incubated Startup)** June 2026 – Present
  - Architected a hybrid RAG platform (pgvector, BM25, Text-to-SQL router), cutting query latency **60s to under 1s**
  - Built a dataset-agnostic ETL pipeline (Spark, Airflow) with a **3-tier** OCR fallback chain across **6** languages
  - Integrated a **9,000+ line** email/scheduling agent into production, validated via CI/CD with RAGAS and TruLens
  - Engineered **3** Intel-optimized RAG modes (INT8-quantized), cutting latency **2-3x** while reaching **92% Recall@10**
- **Machine Learning Intern – Zee Tech and Innovation Centre** December 2025 – January 2026
  - Engineered OmniDoc, a local multimodal RAG engine with BLIP-captioned image retrieval and intent routing
  - Architected a context-aware scene selector, replacing naive top-k selection with budget-optimized ranking
  - Built a full-stack video prototype with user auth, async job processing, and live progress tracking
  - Authored 2 security design specs proposing ML architectures for anomaly detection and content fingerprinting
- **Summer Intern – WESEE, Indian Navy** June 2025 – July 2025
  - Assessed in-house LLMs for naval applications within a secure defense R&D environment
  - Identified bugs, inconsistencies, and logic failures in model outputs for mission-critical reliability
  - Validated Retrieval-Augmented Generation (RAG) pipelines and improved contextual retrieval accuracy
  - Built Artifex.AI, a text-to-image tool integrating Stability AI's SDXL 1.0 API with **sub-5s latency**

Projects

- **AgenticOS:HARNESS – CLI-Based Agentic Operating System**
  - Built a full email pipeline (IMAP/Gmail OAuth2, NLU intent classification, autonomous scheduling) end-to-end
  - Designed a **7-day** collision detection engine with FSM-based thread tracking and single-keystroke HITL approval
  - Cut email processing latency **5-6x (2000ms→300-500ms)** and LLM costs **40%** via response caching
  - Developed a PostgreSQL-backed RAG mode and configured Ollama as a local LLM provider for on-premise reasoning
- **Multiva.AI – AI-Powered Voice Cloning Platform**
  - Engineered an async **11-stage** dubbing pipeline using Whisper, NLLB-200, OpenVoice, and Wav2Lip
  - Built a custom ROI face compositor, cutting post-processing latency **40x (200ms→5ms/frame)**
  - Designed a temporal chunking engine that cut peak VRAM **40%** and lifted success rate to **98%**
  - Optimized STT/translation via CTranslate2 and NLLB-200, achieving **6x** faster transcription
- **S.W.O.R.D – Surveillance for Weapon Observation Using Real-Time Deep Learning**
  - Engineered NMS-free weapon detector using YOLOv26m, reaching **90.1% mAP@0.5** and **82.4% recall**
  - Cut inference time **50ms→7.4ms (135 FPS)** while raising accuracy from **72% to 90.1%** over YOLOv8n baseline
  - Built semi-supervised labeling pipeline with low-light/motion-blur augmentation for CCTV-grade robustness
  - Added temporal tracking to filter false alerts and optimized inference for Jetson Nano/Coral TPU edge deployment

Technical Skills

- **Programming:** Python, Java, C, SQL
- **Frameworks:** LangChain, PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, NumPy, Pandas, Matplotlib, Docling, PySpark, FAISS, Flask, FastAPI, SQLAlchemy
- **Models:** YOLOv8/11/26, Whisper (CTranslate2), Coqui XTTS, OpenVoice, BLIP, Mistral-7B, Gemma4, Qwen 2.5, Wav2Lip, FB-NLLB, BGE-M3 Embeddings, BGE-M3 ReRanker, SentenceTransformers
- **Databases:** PostgreSQL, ChromaDB, SQLite, Elasticsearch, Qdrant, MongoDB
- **Evaluation:** RAGAS, TruLens, BIER, PyTest, CI/CD, InfoVQA
- **Tools:** Git, Docker, OAuth2.0, Cloudflare R2, Ollama, Groq, Google Calendar API, Gradio, Streamlit, Firebase, Supabase, Apache Airflow, Apache Spark, MistralOCR, FFmpeg, Claude

Education

- **B.Tech in Computer Science (AI/ML)** | *IILM University, Greater Noida* 2023 – 2027
- **Class 12** | *Navy Children School, New Delhi* 2023
- **Class 10** | *Navy Children School, Mumbai* 2021

Certifications & Achievements

- **Machine Learning Specialisation** – Stanford University
- **Hackathon Finalist** – Reached the finalist round in 3 of 6 AI/ML hackathons entered
- **DSA** – Solved 150+ questions on LeetCode